

Ann Biju Mariyam

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Results-driven Data Analyst with hands-on experience in data analysis, statistical analysis, data visualization, and dashboard development across retail, e-commerce, and public health domains. Proficient in SQL (JOINS, aggregations, window functions), Python (Pandas, NumPy), Tableau, and Power BI to generate business insights that support data-driven decision-making and stakeholder reporting. Skilled in data cleaning, data validation, data pipeline development, ETL pipelines, time series forecasting, and KPI development. Implemented and evaluated machine learning models using Scikit-learn and TensorFlow with a focus on real-world deployment impact.

PROFESSIONAL EXPERIENCE

Data Analytics Intern, ROGERSOFT TECHNOLOGIES, India

Sep 2024 – Dec 2024

SQL-driven analytics and reporting workflow supporting product performance, customer behavior, and operational KPI tracking across internal business systems.

- **10,000+ records cleaned and validated:** Streamlined data preparation using Python, Pandas, and Excel, improving dataset reliability, reducing reporting inconsistencies, and preparing analysis-ready data for business teams.
- **SQL-based reporting layer:** Queried and transformed product and customer datasets using multi-table JOINS, GROUP BY, aggregations, and conditional filters to create performance metrics for stakeholder reporting.
- **Self-service Power BI dashboards:** Built interactive dashboards with KPI cards, trend lines, bar charts, and filters, enabling teams to monitor operational KPIs, identify trends, and reduce manual reporting dependency.
- **Faster stakeholder reporting:** Converted raw business data into clear performance insights and standardized reporting workflows, reducing manual analysis overhead for cross-functional business teams.

PROJECTS

Sales Revenue Analytics [GitHub](#)

End-to-end sales analytics workflow using Python, SQL, Tableau, and Power BI to diagnose profitability gaps, track revenue trends, and deliver business recommendations from retail transaction data.

- **9,994 transactions analyzed, ~\$2.3M revenue:** Cleaned and explored retail sales data using Python, Pandas, SQL, Matplotlib, and Seaborn, identifying revenue patterns, product-level performance trends, and key profit drivers.
- **Profitability gaps & BI insights delivered:** Pinpointed Technology as the top profit category (\$145K), Tables as the largest loss driver (-\$17.7K), and 30%+ discounts as a margin risk; built Tableau and Power BI dashboards with KPI tracking, monthly revenue trends, and stakeholder-ready recommendations.

E-commerce Customer Analytics & ETL [GitHub](#)

End-to-end ETL and analytics workflow using Python and SQL to unify multi-table e-commerce data and deliver insights across customer experience, logistics, retention, and revenue performance.

- **98,666 orders integrated, ~\$15.84M revenue:** Built an ETL pipeline using Python, Pandas, NumPy, and SQL to combine 6 relational tables into a unified reporting layer for consistent business analysis.
- **Delivery and logistics insights delivered:** Analyzed delivery timelines, freight costs, review scores, and state-level performance, surfacing the link between longer delivery times and lower customer ratings; built Power BI dashboards supporting a prioritized operational improvement roadmap.

COVID-19 Regional Forecasting – Italy(MS Capstone) [GitHub](#)

End-to-end forecasting pipeline using Python to analyze regional clinical epidemiological data and compare statistical and machine learning models for public health resource planning.

- **SARIMA achieved the lowest forecasting error:** Developed and benchmarked Naive Baseline, SARIMA, and Random Forest models using Python, Pandas, and Scikit-learn, with SARIMA improving MAE to 180 compared to Naive MAE of 206.
- **Research-grade validation completed:** Identified and corrected a critical model evaluation flaw through statistical validation and data integrity checks, ensuring reliable forecasting outputs for stakeholder-ready reporting.

SKILLS & CERTIFICATIONS

Skills:

Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn), SQL, R Programming

Analytics: Exploratory Data Analysis, Data Summarization, Statistical Analysis, Statistical Modelling, Hypothesis Testing, Trend Analysis, Data Cleaning, Data Validation, Data Quality, ETL Pipelines, Time Series Forecasting, KPI Analysis, Stakeholder Reporting

Databases: MySQL, Microsoft SQL Server, Relational Database Design

Visualization & Reporting: Power BI (Power Query, DAX, Dashboard, KPI Cards), Tableau, Excel(Pivot Tables, XLOOKUP, Advanced Analysis), Looker

Tools: GitHub, Jupyter Notebook, Google Colab

Machine Learning & Analytics: Supervised ML, Unsupervised ML

Certifications:

Microsoft Power BI Data Analyst Associate (PL-300), Google Cloud Data Analytics.

EDUCATION

UNIVERSITY OF ILLINOIS SPRINGFIELD

MS in Data Analytics (GPA: 3.9/4)

Springfield, IL, USA

May 2026

FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY

BTECH in Computer Science and Engineering

Kerala, India

June 2024